



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

STATISTICS
PAPER 1

6046/1
3 hours

NOVEMBER 2025 SESSION

Additional materials:
Answer booklet
Graph paper
List of Formulae MF 7
Non programmable Scientific calculator
Mathematical Instruments

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer booklet.

Answer **all** questions.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is 120.

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Answer **all** questions.

- 1 State any **one** advantage and any **one** disadvantage of grouping data. [2]
- 2 A vehicle registration system uses a code to register its vehicles for a certain province. The code consists of an ordering of the letters A to C followed by an ordering of the whole number 0 to 9.
Find the number of different codes that can be formed. [2]
- 3 (a) Define the following terms:
 - (i) exhaustive events [2]
 - (ii) mutually exclusive events. [1]
- 4 (b) State any **one** example of an independent event. [1]
- (a) State **one** reason why it is often desirable to take samples. [1]
- (b) Define the term *sampling frame*. [1]
- (c) State any **one** advantage and any **one** disadvantage of quota sampling. [2]
- 5 Name and describe any **two** methods of collecting data. [4]
- 6 (a) State any **three** characteristics of the normal distribution curve. [3]
- (b) Give any **two** life samples with features that follow a normal distribution curve. [2]
- 7 (a) Define the following terms:
 - (i) permutation [1]
 - (ii) combination [1]
- (b) Using factorial notation, evaluate the following:
 - (i) $7P_3$ [1]
 - (ii) ${}^8C_5 \times {}^4C_3$ [2]



- 8 A committee of 3 men and 2 women is to be formed from a group of 4 men and 4 women.

Find the number of possible committees that can be formed if

- (a) there is no restriction of who should be in the committee, [2]
 (b) there should be a particular woman and a particular man in the committee. [3]

- 9 (a) Distinguish between a Bernouli and a Binomial distribution. [2]

- (b) Given $X \sim \text{Bin} \left(7; \frac{1}{2} \right)$,
 find $P(X = 5)$ [2]

- (c) Given $X \sim \text{Bin} \left(10; \frac{1}{4} \right)$,
 find $P(X > 3)$ [2]

- 10 Every year A level learners in a particular province write a common midyear examination. Over a period of time it has been established that the marks can be modelled by a normal distribution with mean 66 and standard deviation 3.

This year it was noted that the learners from that particular province did not perform as expected. The marks of a randomly selected sample of 20 learners came up with a mean mark was 64.8 Test at 5% level of significance whether the learners did not perform well. [7]

- 11 A boarding master at a certain high school conducted a survey on sleeping times of boys. A random sample of 15 form 4 boys was used. The number of hours of sleep are shown below.

4.6;	6.9;	7.0;	6.5;	8.1;	9.0;
9.3;	8.1;	7.1;	5.9;	6.8;	7.3;
7.1	7.7	7.7			

- (a) (i) Find a reason why this is an unsatisfactory sample for the purpose of survey. [1]
 (ii) State a population for which this sample would be satisfactory. [1]



- (b) Calculate the unbiased estimates of the
- (i) mean, [2]
 - (ii) variance. [3]
- 12 (a) Give any **two** advantages and any **two** disadvantages of using experiments as a method of collecting data. [4]
- (b) (i) Distinguish between *discrete* and *continuous* data. [2]
- (ii) Give an example for each of (i). [2]
- 13 The masses of apples from a given tree are normally distributed with mean 600 g and standard deviation 5 g.
- Find the probability that a randomly selected apple weighs
- (a) less than 590 g, [3]
 - (b) more than 608 g, [3]
 - (c) between 595 g and 599 g. [3]
- 14 (a) State any **two** conditions for a Binomial distribution. [2]
- (b) A gunman is asked to take shots at a target and the number of hits are counted. The probability of hitting the target with a single shot is 0,7.
- (i) Find the probability that in ten attempts the target is hit at most four times. [5]
 - (ii) Explain why a Binomial distribution may not be a suitable model to use in this situation. [1]
- 15 (a) Define the following terms:
- (i) mean [1]
 - (ii) mode [1]
 - (iii) median [1]
- (b) State any **one** advantage and any **one** disadvantage of each of the terms in (a). [6]



- 16 X and Y are independent normal random variables with means and variances as shown in the table below.

	Mean	Variance
X	20	6
Y	8	2

- (a) Find the distribution of,
- (i) $X + 5$ [3]
- (ii) $4X + Y$ [5]
- (b) Calculate $P(4X + Y) \geq 90$ [3]
- 17 A shop manager claims that the average time for standing in the queue before being served is 5.2 minutes. Ten customers were asked how long they stood in the queue before they got served. The following were their times in minutes:
- 5.8 4.9 5.3 6.1 6.2 4.7 5.4 5.6 5.0 5.7
- Is there evidence, at the 5% level, that the managers claim is true? [11]
- 18 Ten pairs of observations on the variables x and y are given below.

Observation	1	2	3	4	5	6	7	8	9	10
x	9	10	12	6	9	8	5	9	4	6
y	10	11	10	9	7	8	7	8	7	4

- (a) Show the data on a scatter diagram. [3]
- (b) Calculate the equation of the estimated regression line of Y on X and draw it on the scatter diagram. [6]
- (c) Calculate Pearson's product moment correlation coefficient and comment on it. [3]
- (d) Find the coefficient of determination. [2]





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STATISTICS
PAPER 2

6046/2
3 hours

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Additional materials:
Answer Booklet
List of Formulae MF 7
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Mathematical Instruments

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer booklet.

Answer **all** questions in **Section A** and any **five** from **Section B**.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

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Section A (40 marks)

Answer all questions in this section.

- 1 The allowances claimed by 28 delegates attending a workshop are as follows:

34	39	40	14	60	24	30
38	41	44	27	43	33	35
38	31	63	34	39	50	55
56	33	38	48	47	54	41

- (a) Construct a stem and leaf diagram for the above data. [4]
- (b) State any **one** advantage of using a stem and leaf diagram for representation of data. [1]
- 2 The heights of 20 boys in a class were measured and recorded, to the nearest centimetre, as follows:

132	135	138	131	137	135
139	129	135	133	138	140
131	131	136	142	141	139
136	141				

- (a) State the (i) mode, [1]
(ii) range. [1]
- (b) Calculate the (i) mean, [2]
(ii) standard deviation. [3]
- 3 The heights of pupils at a certain high school are normally distributed such that 40% of them are greater than 1.5 metres and 30% are greater than 1.64 metres. Estimate the mean and standard deviation of their heights. [8]
- 4 (a) Distinguish between a *sample* and a *population*. [2]
- (b) Distinguish between *probability sampling* and *non-probability sampling*. [2]
- (c) A sports director requires to obtain the views of the learners about taking wrestling as a sporting activity. This would be done by means of a small sample.

Describe briefly how you would select such a sample using

- (i) stratified random sampling, [2]
- (ii) simple random sampling. [2]



- (d) State with a reason, which of the two sampling methods in (c) would be considered as the most appropriate for this situation. [2]

5 (a) Define the following terms

- (i) null hypothesis, [1]
 (ii) alternative hypothesis, [1]
 (iii) test statistic. [1]

- (b) After complaints from consumers that their products were under-weight, the production manager wanted to establish if the consumers' claim was correct. He randomly picked 10 packets from the conveyer belt. The company claims that their product has a mean of 10 kg. The following are the masses of the sample.

10.5 kg	9.8 kg	11 kg	9.7 kg	8.9 kg
10 kg	9.6 kg	9.3 kg	9.9 kg	9.2 kg

Assuming that the masses follow a normal distribution, test at 10% significance level, whether the consumers' claim that the packets are underweight is correct? [7]



Section B (80 marks)

Answer any **five** questions from this section.

Each question carries 16 marks.

- 6 (a) The continuous random variable X has an exponential distribution with mean μ .

Given that $P(X < 1) = \frac{1}{4}$.

Find μ . [5]

- (b) The continuous random variable X has an exponential distribution with mean 3.

Find

(i) the probability density function of X , [2]

(ii) the cumulative distribution function of X , [2]

(iii) $P(X < 3)$, [2]

(iv) $P(X > 4)$, [2]

(v) $P(3 < X < 4)$. [3]

- 7 In a public opinion poll, one thousand randomly chosen electors were asked whether they would vote for the party A, at the next election and 357 replied yes.

(a) Find a 95% confidence interval for the proportion, P , of the population who would answer yes to the same question. [6]

(b) Twenty similar polls are taken and the 95% confidence interval is determined for each poll.

State the expected number of those intervals which will enclose the true value of P . [2]

(c) The leader of party A believes that the true value of P is 0.4.

Test at the 8% level whether the leader is over estimating their support. [8]



- 8 (a) Distinguish between a *Type I error* and a *Type II error*. [2]
- (b) An electric company claims that its solar light lasts up to an average of 1000hrs after having been fully charged. A consumer council thinks this is an over estimate and tests a random sample of 64 bulbs. The results are summarised as follows:

$$\sum x = 63\,910.4$$

$$\sum x^2 = 63\,824\,061$$

- (i) Calculate the sample mean. [1]
- (ii) Calculate an unbiased estimate of the standard deviation. [3]
- (iii) Test at 5% level of significance whether the company is over estimating the average life time of the solar light. [8]
- (iv) State any **one** assumption made. [1]
- (v) Find the probability of making a *Type I error*. [1]
- 9 The continuous random variable X is given by probability function.

$$f(X) = ke^{-\frac{1}{3}X}, X > 0$$

- (a) Using integration, show that $k = \frac{1}{3}$. [3]
- (b) Find $P(1 < X < 3)$. [3]
- (c) Using integration by parts show that $E(X) = 3$. [8]
- (d) Calculate $\text{Var}(X)$. [2]
- 10 The following table shows the profit in thousands of dollars made by a company over a period of 12 years.

Year	1	2	3	4	5	6	7	8	9	10	11	12
Profit	62	58	74	102	95	90	104	135	126	120	134	156

- (a) Draw a time series graph. [3]
- (b) Use a four yearly moving average to obtain the trend line. [9]
- (c) Use the trend line to forecast the profit in the 13th year. [4]



- 11 A researcher was studying the population of a city and its area. In a survey, a random sample of 12 cities with their area, X , in thousands of square kilometres and population, Y , in thousands were noted and the results are given below.

City	A	B	C	D	E	F	G	H	L	J	K	L
Area X in thousands of Km^2	61	23	8	14	42	34	32	31	41	25	20	50
Population Y , in thousands	49	12	3	4	28	27	20	20	34	15	16	40

- (a) Plot the data on a scatter diagram. [4]
- (b) (i) Calculate the regression equation in the form $y = a + bx$. [5]
- (ii) Draw the line of best fit for this data. [2]
- (c) Estimate the population of a city with an area of 60 000 km^2 . [2]
- (d) (i) Calculate the product moment correlation coefficient.
- (ii) Comment on the answer. [3]
- 12 The length of a line that can be written by two types of pens, A and B, follow normal distributions with parameters given in the table below.

Type	Mean	Standard deviation
A	5 km	0.5 km
B	4.8 km	0.6 km

Find the probability that

- (a) a type A pen can write a line greater than 5.3 km, [2]
- (b) a type B pen can write a line greater than a type A pen, [4]
- (c) 3 type A pens will write lines with a total length less than 14 km, [5]
- (d) the total length written by 3 type A pens is greater than the total length written by 4 type B pens. [5]

